

CUSTOMER JOURNEY ANALYSIS

Conversion funnel, behavioural patterns, and optimisation priorities



Funnel Analysis

EDA

Excel

Excel Solver

Python

DESCRIPTION

Customer Journey Funnel Optimisation and Resource Prioritisation

Executive Summary

Approximately 20.2% of users completed a purchase. The largest point of friction occurred between the product page and cart, where almost 60% of users dropped out.

The pattern was consistent across continents, suggesting a structural experience problem rather than a regional one. Afternoon users showed the strongest conversion efficiency, while evening traffic was highest in volume but weaker in conversion.

20.2%

PURCHASE COMPLETION

59.9%

PRODUCT-TO-CART DROP-OFF

60-70%

PREPARATION TIME SAVED

Problem

Despite strong traffic volume, many users failed to complete a purchase. The analysis aimed to identify where users disengaged, examine likely causes, and recommend improvements that support both customer experience and conversion.

Analytical Approach

Data reconstruction

- Processed event-level session data to rebuild the full journey from landing page to purchase confirmation.
- Validated stage order, removed inconsistencies, and aggregated user-level outcomes.

Comparative analysis

- Segmented users by device type, geography, referral source, and time of day.
- Compared stage retention and conversion efficiency across groups to identify repeatable patterns.

Technical Implementation

Python

- Converted source CSV files to XLSX
- Cleaned and validated event data
- Executed SQL-style aggregations and prepared analysis tables

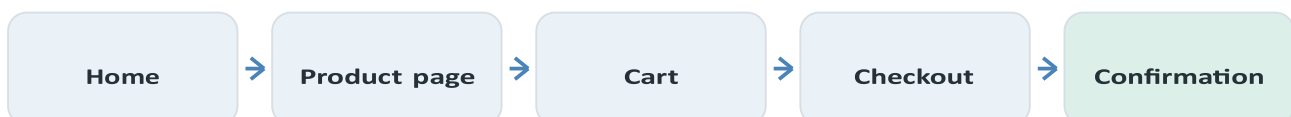
Excel

- Built pivot tables from prepared datasets
- Created charts and presentation-ready visualisations

60-70% less preparation time

The automated workflow made the analysis faster to reproduce and easier to iterate.

Funnel Overview



Funnel Retention

Users retained at each funnel stage

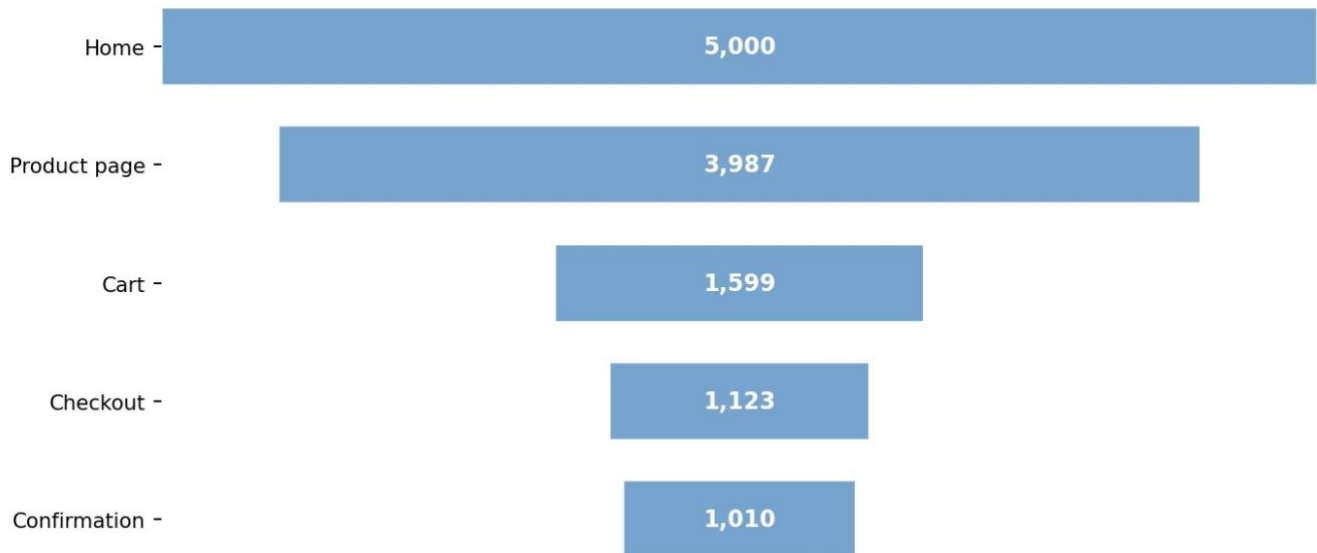


Figure 1. Users retained at each stage of the funnel.

Stage	Users retained	Stage conversion	Drop-off
Home	5,000	-	-
Product page	3,987	79.7%	20.3%
Cart	1,599	40.1%	59.9%
Checkout	1,123	70.2%	29.8%
Confirmation	1,010	89.9%	10.1%

What this shows

The funnel is relatively healthy at entry and checkout. The major break occurs during product evaluation: users reach the product page, but many do not add an item to the cart.

Funnel Drop-Off Analysis

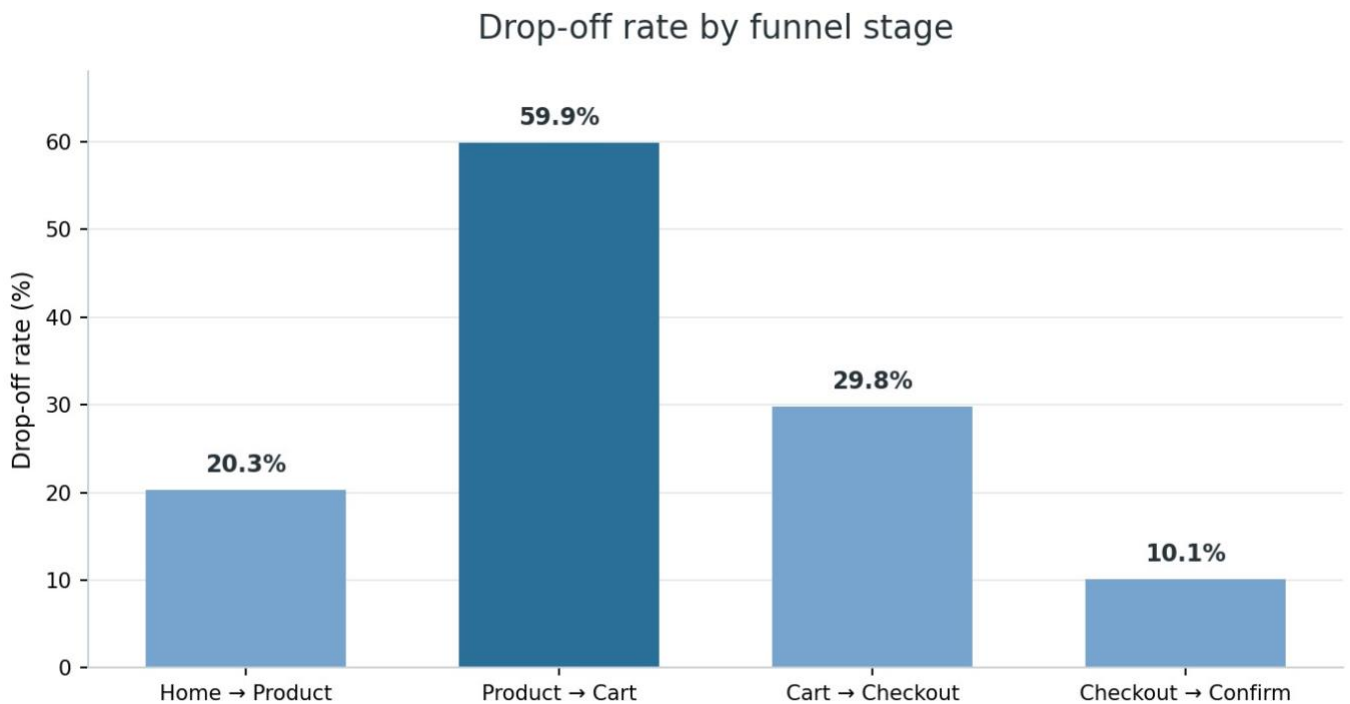


Figure 2. Drop-off rates between consecutive funnel stages.

Key Findings

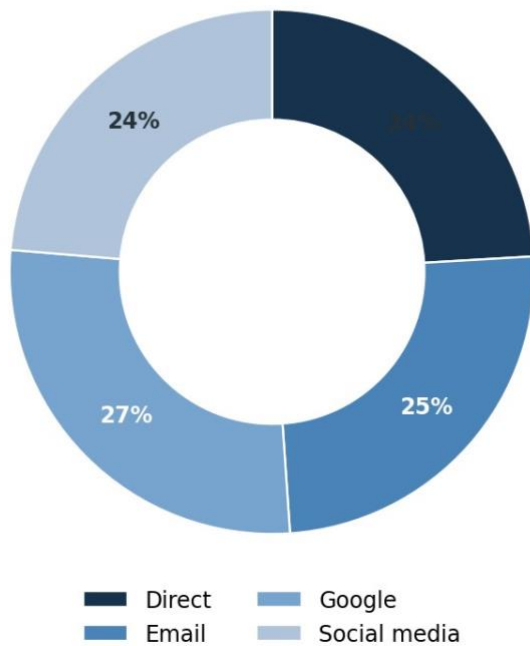
- 20.2% of all users reached purchase confirmation.
- The product page to cart transition had the highest drop-off at 59.9%.
- Drop-off from cart to checkout was moderate at 29.8%.
- Once users entered checkout, completion was comparatively strong.

Interpretation

Users showed browsing intent but hesitated during product evaluation. This points to friction in pricing clarity, perceived value, trust signals, product information, or usability rather than a checkout-only problem.

Referral Source Analysis

Confirmed purchases by referral source



Source	Purchases	Share
Direct	243	24.06%
Email	251	24.85%
Google	277	27.43%
Social media	239	23.66%

Figure 3. Distribution of confirmed purchases by referral source.

Insights

- Google contributed the largest share of completed purchases at 27.43%, followed by Email at 24.85%.
- The distribution was balanced; no single channel dominated completed purchases.
- Because volume was broadly spread, improving conversion efficiency is likely more valuable than simply increasing traffic.

Resource prioritisation implication

The evidence supports prioritising product experience and development improvements before allocating additional budget to traffic acquisition.

Geographic Patterns

Cross-region comparison



Qualitative summary based on the report's observed regional pattern; no region-specific values were presented.

Insights

- Drop-off patterns were broadly consistent across continents.
- No single region disproportionately underperformed.
- The shared shape of the funnel suggests a structural product or experience issue.

Implication

Global product-page and usability improvements should be prioritised before region-specific redesigns. Regional experiments can follow once the core experience has been strengthened.

Optimisation Priority

1

Product experience

Clarify pricing, value, trust signals, and product information.

2

Mobile usability

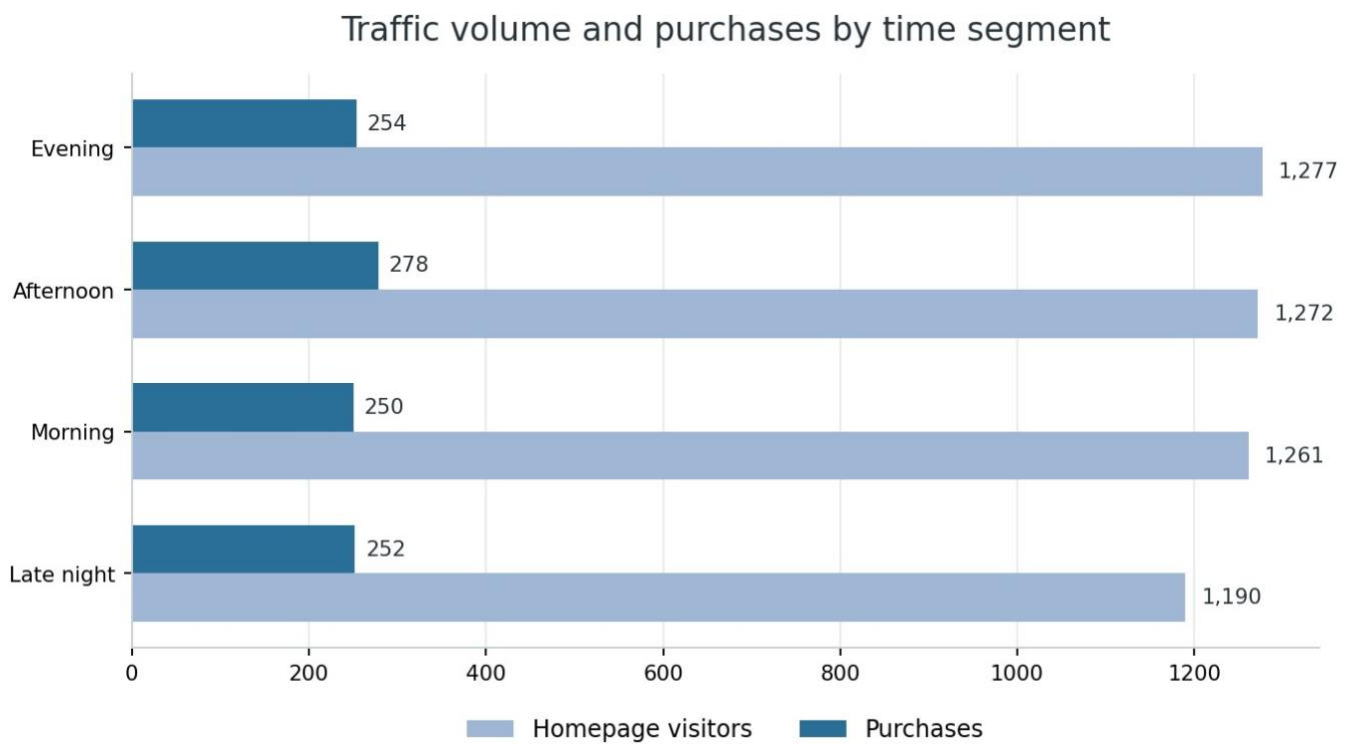
Reduce friction at the product and cart stages.

3

Campaign timing

Use high-intent time windows for promotions and nudges.

Time-of-Day Analysis



Time segment	Visitors	Purchases	Conversion
Evening	1,277	254	19.9%
Afternoon	1,272	278	21.9%
Morning	1,261	250	19.8%
Late night	1,190	252	21.2%

Key Behavioural Insights

- Evening sessions generated the highest traffic volume, but conversion was comparatively weak.
- Afternoon users had the highest observed conversion rate at about 21.9%.
- Late-night traffic was smaller but converted efficiently at about 21.2%.

Opportunity

Align promotions, reminders, and engagement nudges with high-intent periods. Test afternoon and late-night timing rather than assuming the highest-volume period will perform best.

Key Outcomes and Impact

Identified the primary funnel bottleneck at the product evaluation stage.

Confirmed that conversion challenges were structural rather than regional.

Revealed meaningful time-based differences in conversion efficiency.

Reduced preparation time by 60-70% through workflow automation.

Actionable Recommendations

1

Optimise product pages

Use clearer pricing, stronger value messaging, better product information, and visible trust signals.

2

Improve mobile usability

Reduce friction at product and cart stages, especially around calls to action and form entry.

3

Align promotions with intent

Schedule campaigns and nudges around the strongest observed conversion windows.

4

Validate with controlled tests

Run A/B tests and measure movement in stage conversion, not only total purchases.

What I'd Do Next

- Implement an automated funnel-monitoring dashboard with stage-level alerts.
- Run A/B tests on product-page layout, pricing presentation, and trust messaging.
- Track conversion changes after time-based campaign adjustments.
- Add device-level analysis to isolate mobile-specific friction.